6 **(a)** Electrons are emitted from a metal surface when light of a particular wavelength is incident on the surface. Explain why the emitted electrons have a range of values of kinetic energy below a maximum value.

.....

.....

.....

..... [2]

(b) An evacuated tube contains two parallel metal electrodes, one of which is an emitter of electrons and the other a collector. The emitter has a work function energy of 2.0 eV and is illuminated with electromagnetic radiation of wavelength 410 nm. The area illuminated on the emitter is 24 mm^2. The potential difference *V* between the collector and the emitter is adjusted, and the photocurrent *I* is measured. Fig. 6.1 shows the variation of *I* with *V*.

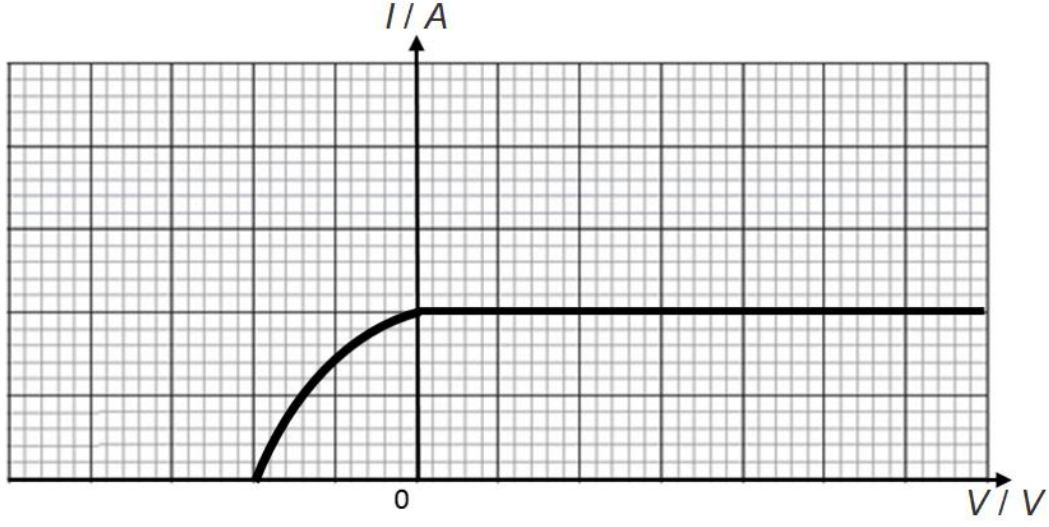

Fig 6.1

On Fig 6.1, sketch graphs to show the variation of *I* with *V* when the following changes are made to the original setup.

(i) The wavelength of the electromagnetic radiation is kept constant but its intensity is reduced by half. Label the graph **(i)**. [2]

(ii) The wavelength of the electromagnetic radiation is decreased but its intensity is kept constant. Label the graph **(ii)**. [2]

(iii) At a particular setting, a photocurrent of 4.8×10^{-10} A was observed.

1. Determine the rate of emission of photoelectrons.

rate of emission = ……. s^{-1} [1]

2. Hence, determine the intensity of the light source, assuming that 1 in 2500 photons succeeds in ejecting an electron from the surface.

intensity = ……. W m^{-2} [3]